P-D-455D
January 14, 1975
SUPERSEDING
Fed. Spec. P-D-455C
October 8, 1970

FEDERAL SPECIFICATION

DISKS, ABRASIVE, ALUMINUM-OXIDE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers aluminum-oxide coated disks commonly used in portable sanding machines for polishing, finishing, grinding and removal of rust or scale.

1.2 Classification.

1.2.1 Classes and grit sizes. Disks shall be of the following types and grit sizes, as specified (see 6.2):

Class 1 - Aluminum-oxide, closed coat.
Grit Size: 180-150-120-100-80-60-50-36-24-16.
Class 2 - Aluminum-oxide, open coat.
Grit Size: 36-24-16.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-3816 - Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

FSC 5345

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Commerce Product Standard:

PS 8 - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies shaould be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

American Society for Testing and Materials (ASTM) Standard:

D828 - Tensile Breaking Strength of Paper and Paperboard.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Abrasive Coated disks shall be of the design, dimensions, and tolerances shown in table I. The measurements shall be made after conditioning at 70°F plus or minus 2°F and 50 percent plus or minus 2 percent relative humidity for at least 24 hours.

TABLE I. Design, dimensions, and tolerances

Design	1 Outside diameter Inches	2 Arbor hole diameter Inch
With slits or slots.	9-1/8, 7, 6	7/8
Plain - no slits or slots.	9-1/8, 7, 6, 5, 4	1/2, 5/8, 7/8

1 Tolerance of plus or minus 1/16 inch will be permitted.
2 Tolerance of plus 1/16 inch will be permitted.

3.2 Slits or slots. A slit shall be considered as an incision having no appreciable width. A slot shall be considered as a long narrow opening having an appreciable width. Slits or Slots shall be two or four in number as specified (see 6.2). Slits or slots have an over-all length of 1-1/16", shall extend radially from the arbor hole, and shall be equally spaced. Slots shall be approximately 1/16" to 1/8" in width with the outer end of the slot terminated in an arc having a radius approximately 1/32" to 1/16", or in a circular opening having a diameter less than 1/4".

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3.3 Grading, control and tolerances. The grading control and tolerances of the abrasive grain shall be in accordance with Product Standard PS8, Grading of Abrasive Grain on Coated Abrasive Products.

3.4 Vulcanized fiber back.

3.4.1 Thickness. The thickness of the vulcanized fiber backing shall be not less than 0.023 for grit size 80 and finer and not less than 0.028 for grit sizes 60 and coarser and not more than 0.042 inch when conditioned at 70° ± 2°F. and 50 ± 2 percent relative humidity for 48 hours.

3.4.2 Breaking strength. The average breaking strength of the coated abrasive disks shall be not less than 350 pounds per inch of width in the machine direction and 180 pounds in the transverse direction of the fiber (see 4.4.2).

3.4.3 Adhesion between plies. When more than one ply is used, the adhesion between the plies shall be not less than 900 g. (see 4.4.3).

3.5 Class 1 and class 2, synthetic resin bond, vulcanized fiber back. The backing shall produce a condition of flexibility and strength, and shall be satisfactory for the operation of the disks in grinding rough surfaces at high speeds.

3.5.1 Class 1, aluminum-oxide, closed coat.

3.5.1.1 Adhesive. The adhesive shall be of good quality, nonhydroscopic, made from synthetic resins, and shall be resistant to severe heat and humidity conditions.

3.5.1.2 Abrasive. The abrasive shall conform to 3.5.2.2.

3.5.1.3 Closed coat. The abrasive grit shall be applied in such a manner that the grit coating shall be dense and completely cover the surface of the backing.

3.5.2 Class 2, aluminum-oxide, open coat.

3.5.2.1 Adhesive. The adhesive shall be of good quality, nonhydroscopic, made from synthetic resins and shall be resistant to severe heat and humidity conditions.

3.5.2.2 Abrasive. The abrasive shall be of good quality Aluminum-oxide, free from extraneous material and entirely suitable for grinding on materials of high tensile strength at high speeds.

3.5.2.3 Open coat. The abrasive grit shall be applied in such a manner that approximately 50 percent of the backing will be covered, thus leaving open spaces between the grains of the abrasive.

3.6 Marking. The abrasive coated disks shall be marked in a plain and legible manner with the manufacturer's name or trademark, kind of abrasive (manufacturer's trade name) class of coating, and grit size.

3.7 Workmanship. The abrasive coated disks shall be free from cracks, uneven distribution of grit, frayed edges, or other imperfections which may impair the serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all disks of the same type, class, and grit size submitted for inspection at the same time.

4.2.2 Visual examination of end item. Sampling for this examination shall be according to level S-2 in MIL-STD-105. The Acceptable Quality Levels (AQL's) for defects listed in table II shall be 1.5 percent defective for major defects and 4.5 percent defective for total defects.

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4.2.3 Sampling for tests. Sampling for tests shall be in accordance with level S-2 in MIL-STD-105, the AQL for tests shall be 4.0 on the basis of percent defective.

4.3 Examination.

4.3.1 End item. Disks selected in accordance with 4.2.2 shall be examined for defects listed in table II.

TABLE II. Classification of defects, end item

Defects	Major	Minor
Type, class, size, or grit not as specified.	X	
Frayed or unevenly cut edges.	X	
Mounting hole not centrally located (where applicable).	X	
Broken, or folded.	X	
Voids in coating (closed coat only).		X
Loose grit.	X	
Contains extraneous matter.	X	
Damage or defects affecting function or serviceability.	X	
Damage or defects not affecting function or serviciability.		X

4.4 Test procedures.

4.4.1 Grading for grain size. Grading to determine the grain (grit) size, including recovery of grain from the disks, preparation of grain for grading, and procedures shall be in accordance with the grading of aluminum-oxide grain as applicable as included in Product Standard PS8 Grading of Abrasive Grain on Coated Abrasive Products.

4.4.1.1 Certification grading. Grading tests specified in 4.4.1 may be omitted in instance where the contractor produces certification in writing from the actual manufacturer that such abrasive grain was graded in accordance with Product Standard PS8.

4.4.2 Breaking-strength. The breaking strength (see 3.4.2) of the coated abrasive disks shall be determined by Method ASTM D828, except that the jaws shall be five inches apart and five specimens each, either one-half inch or one inch wide, shall be taken longtudinally* and transversely* from the sample disks. The samples shall be conditioned for at least 48 hours at 70°F. ± 2°F. and 50% ± 2% relative humidity. The average breaking strength shall be computed from at least three specimens.

*The approximate machine and transverse directions shall be established by visual and phusical inspection of the sample disks.

4.4.3 Adhesion. The adhesion between plies of uncoated vulcanized fiber shall be conducted on a one-half inch strip cut transversely from the fiber. The adhesion resistance shall be measured on a ply adhesion testing machine designed for this purpose, after the half inch strip has been conditioned for 48 hours at 70° ± 2°F. and 50 ± 2 percent relative humidity (see 3.4.3).

4.5 Inspection of preparation for delivery. The preservation, packaging, packing and marking of the disks shall be examined to determine compliance with the requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The disks shall be packaged in accordance with the level A requirements of MIL-A-3816.

5.1.2 Level B. Twenty-five disks, of one stock number, shall be packaged in accordance with the level B requirements of MIL-S-3816.

5.1.3 Level C. Disks shall be packaged in accordance with the level C requirements of MIL-A-3816.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unit packages shall be packed in accordance with the level A requirements of MIL-A-3816.

5.2.2 Level B. Ten unit packages (250 disks) shall be packed in accordance with level B requirements of MIL-A-3816.

5.2.3 Level C. Unit packages shall be packed in accordance with the level C requirements of MIL-A-3816.

5.3 Marking. In addition to any special marking required by the contract or order, the unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129 or Fed. Std. No. 123, as applicable.

6. NOTES

6.1 Intended use. Abrasive disks covered by this specification are intended for use in specific operations within the limits of its applicability as follows:

(a) Abrasive:
(1) Aluminum-oxide is used in majority of grinding or finishing operations on ferrous metals.

(b) Bond:
(1) Resin bond is best suited for severe metal working operations where grinding heat is a factor or in operations where a lubricant, such as water, is required.

(c) Coating (abrasive):
(1) Open coat disks are generally used for grinding soft materials which display a definite tendency to "load" the disk such as aluminum, fill solder and paint.

(2) Closed coat disks are generally used for grinding harder materials when particles are relatively small and do not have a tendency to "weld" together.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class and grit size of disks (see 1.2).
(c) Design and size (see 3.1).
(d) Whether two or four slots are required (for 7/8 in diameter only) (see 3.2).
(e) Size of lot if different from 4.2.1.
(f) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

MILITARY INTEREST:

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Navy - Ships
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Preparing activity:

GSA - FSS

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